Case 02: Transonic Whitcomb Supercritical Airfoil

Case Details

The objective of the simulation is to validate the Whitcomb (Integral) airfoil's experimental data given in [1] with low computational cost for fast but accurate results.

- Reference: https://ntrs.nasa.gov/citations/19900007394
- Airfoil: Whitcomb supercritical (integral) airfoil
- Mach Number: 0.8
- Reynolds Number: 3.0×10^7
- Angle of Attack: 2°

How to Run

1. Select whitcomb airfoil from the airfoil directory
2. Enter the desired number of coordinates if required and output directory.
3. Click on "Analyze" to obtain the accurate parameterization method and its respective parameters
4. Go to SU2 Settings → Meshing & Conditions
5. Enter Mach, Reynolds number and enter 2 in minimum AoA with 0 step.
6. Select "Load recommended settings"
7. Choose the type of mesh and target y+ value
8. Select "Generate Hybrid Mesh"
9. Go to SU2 Settings → Solver Setup & Run
10. Modify settings as desired or select "Run SU2 Analysis" (validated settings have been provided as default and are given here in the next section)
11. After convergence, the output or post-processed contours will be displayed in the SU2 Settings → Results tab and saved in the output directory.

Solver Settings

Given below are the solver settings validated for this case and are provided by default for this regime. This case is run using unsteady parameters:

Setting	**File**
NUM_METHOD_GRAD	WEIGHTED_LEAST_SQUARES
CONV_NUM_METHOD_FLOW	JST
SLOPE_LIMITER_FLOW	VENKATAKRISHNAN
LINEAR_SOLVER	FGMRES
LINEAR_SOLVER_PREC	ILU
KIND_TURB_MODEL	SA
CFL_NUMBER	0.5
CONV_FIELD	RMS_DENSITY
Min log10 Residual	-6
TIME STEPPING	YES

Table 1: Solver settings used

Result

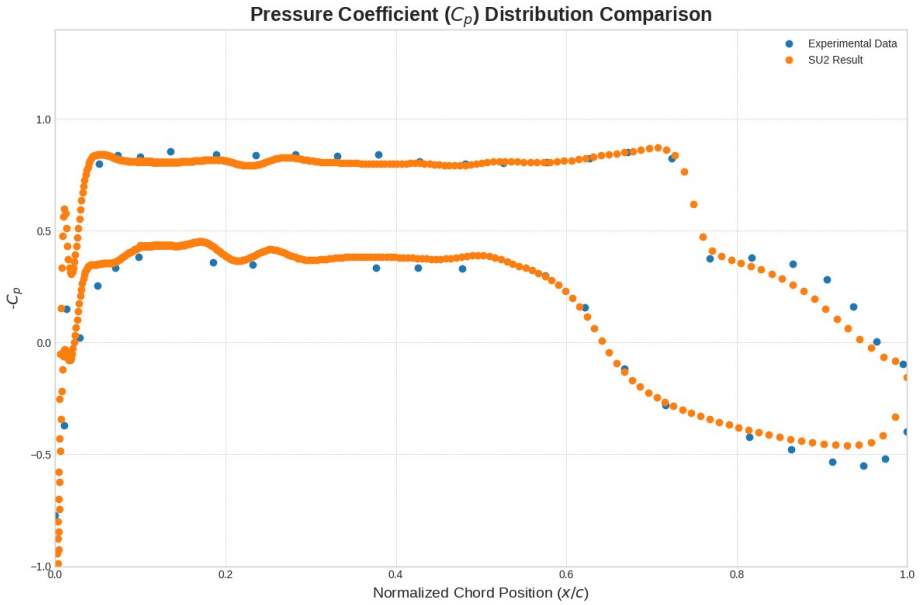


Figure 1: Comparison between coefficient of pressure values between experimental (blue) and FALCON result (orange)

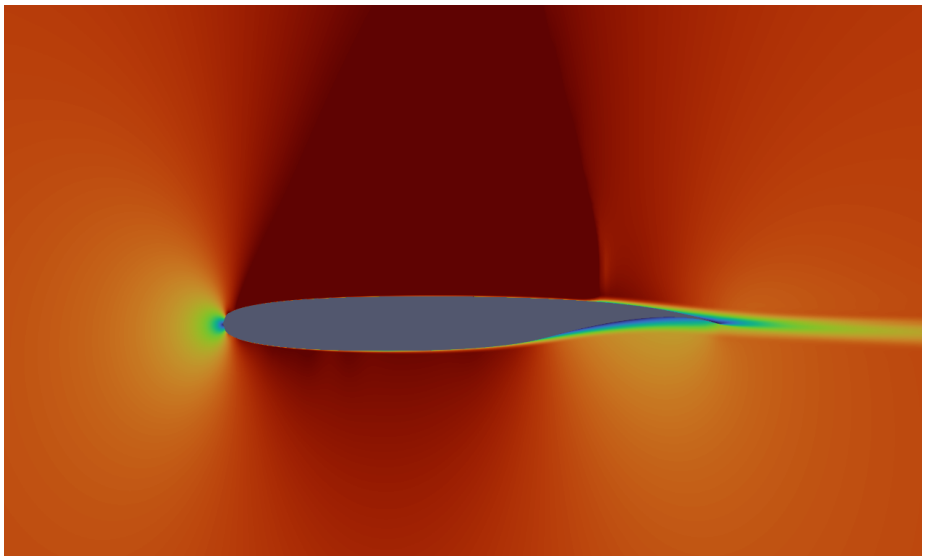


Figure 2: Velocity field as visualized in ParaView

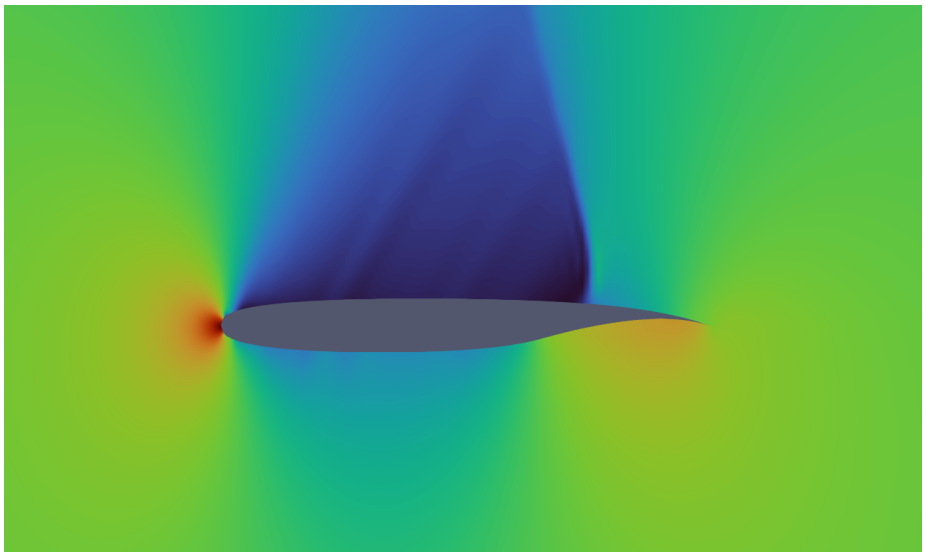


Figure 3: Pressure field as visualized in ParaView

References

[1] C. D. Harris, "NASA supercritical airfoils: A matrix of family-related airfoils," National Aeronautics and Space Administration, Langley Research Center, Hampton, VA, NASA Technical Paper 2969, Mar. 1990. [Online]. Available: `https://nasa.gov`